

Teaching Exam

Subject – Computer Science

Topic – DBMS



Lecture No.- 1

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Topics to be Covered

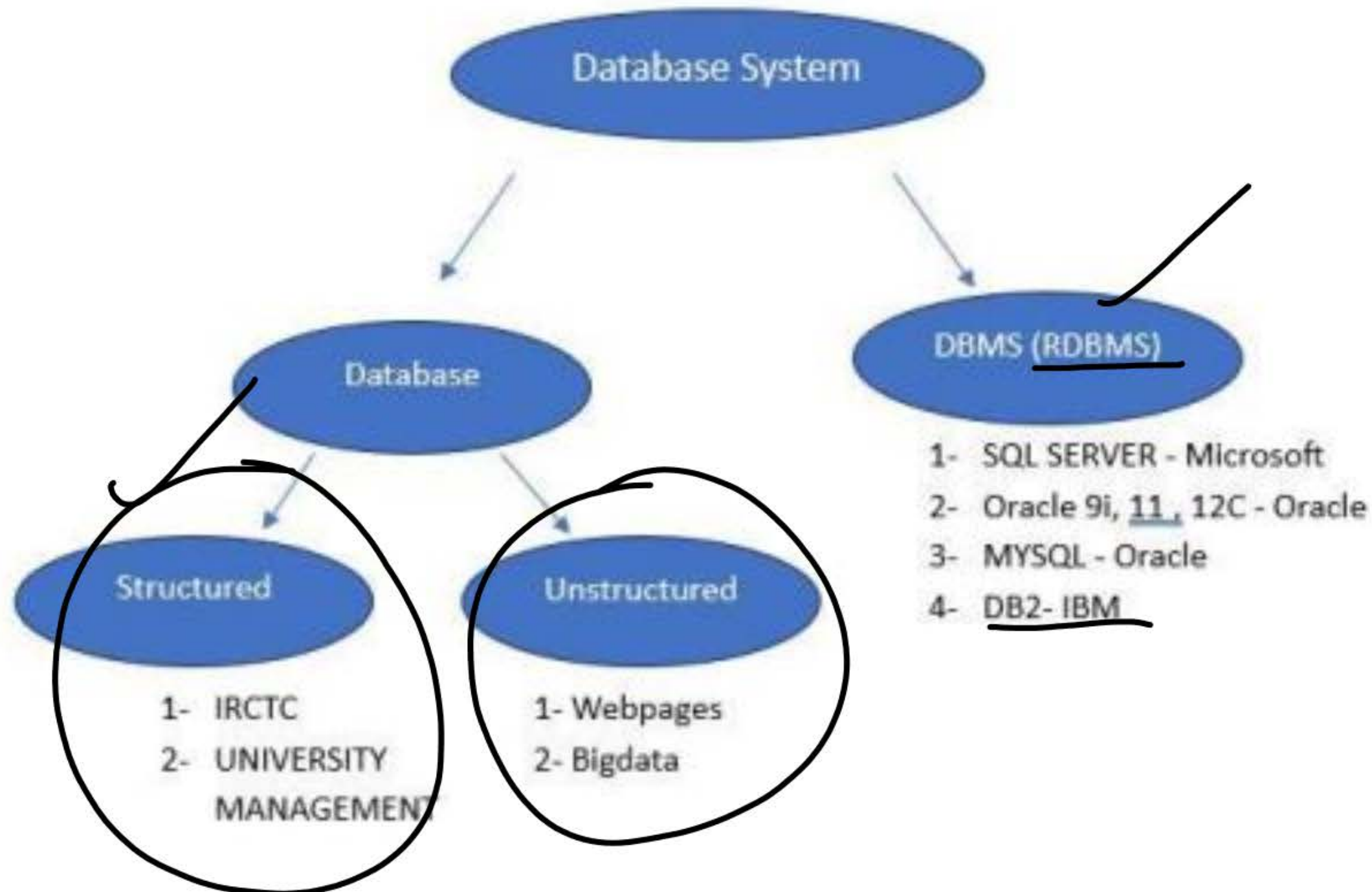
Topic



- 1:- Introduction to DBMS
- 2:- Types of DBMS
- 3:- Difference between Flat file & RDBMS
- 4:- Limitation of Flat File Database
- 5:- Database vs File System
- 6:- 2 –Tier Architecture
- 7:- Advantage and Disadvantage of 2-Tier
- 8:- 3-Tier Architecture
- 9:- Advantage and Disadvantage of 3-Tier



Topic : Introduction to DBMS



Topic : Introduction to DBMS

History

The Founder of the Modern Database is Considered to be Edgar Frank Codd who developed the Relational Model in 1970.

Database System

A Database System or Database Organized Collection of Structured information or data , typically stored Electronically in a Computer System at a Central Location.

Topic : Introduction to DBMS

Database Management System (DBMS)

A Database Management System is Organized Collection of Structured information in under Software, where we can Store, Retrieve and Manipulate our Data.

Ex-

Oracle(9i, 11, 12C), DB2 (IBM), SQL Server (Microsoft), MYSQL (Oracle) and many others.

Topic : Types of DBMS

There are two types DBMS :-

FLAT-FILE:- A Flat-File Database is a Simple, Basic Storage Structure that Stores all data with in a Single Table or File , Without any Relationships between records.

Ex- CSV File, Excel File etc.

RDBMS:- RDBMS Stands for Relational Database Management Systems. It is Program that allow us to Create , Delete and update a relationship database (Inter-related table).

A Relational Database is system that store and retrieve data in a tabular format organized rows and columns.

EX- MYSQL , Oracle (9i,11,12C) etc.

Topic : Limitation of Flat File Database

1- Data Redundancy:- It refers to the Storage of the same data Multiple times.

Patient Id	Name	DOB	Gender	Phone	Doctor Id	Doctor	Visits	Room
134	Javed	4-Jul-2003	Male	7876453	01	Dr. Harpreet	7	03
178	David	8-Feb-1997	Male	8635467	02	Dr. Nishaat	9	06
198	Lata	18-Dec-1989	Female	7498735	01	Dr. Harpreet	7	03
210	Jagpreet	29-Apr-1993	Male	7943521	01	Dr. Harpreet	7	03
258	Anu	8-Feb-1997	Female	8367242	02	Dr. Nishaat	9	06

Duplicate (pointing to the first and third rows)

Duplicate (pointing to the second and fifth rows)

2- Data Inconsistency:- Multiple Mismatched Copies of same data is called Data

Inconsistency.

Patient Id	Name	DOB	Gender	Phone	Doctor Id	Doctor	Visits	Room
134	Javed	4-Jul-2003	Male	7876453	01	Dr. Mohan	7	03
178	David	8-Feb-1997	Male	8635467	02	Dr. Nishaat	9	06
198	Lata	18-Dec-1989	Female	7498735	01	Dr. Mohan	7	03
210	Jagpreet	29-Apr-1993	Male	7943521	01	Dr. Harpreet	7	03
258	Anu	8-Feb-1997	Female	8367242	02	Dr. Nishaat	9	06

Multiple copies of Doctor id 01, but with different names (data inconsistency) (pointing to the first, third, and fourth rows)

DATA INCONSISTENCY

Topic : Limitation of Flat File Database

- 1- Data Isolation:-** It refers to a Situation where data of one file can not be mapped to other related file in the absence of links or mappings or common formats.
- 2- Data Sharing Security/Control Issues:-** With File based systems , there is no Provision to have Control over data access. All the data is available to all types of Users.

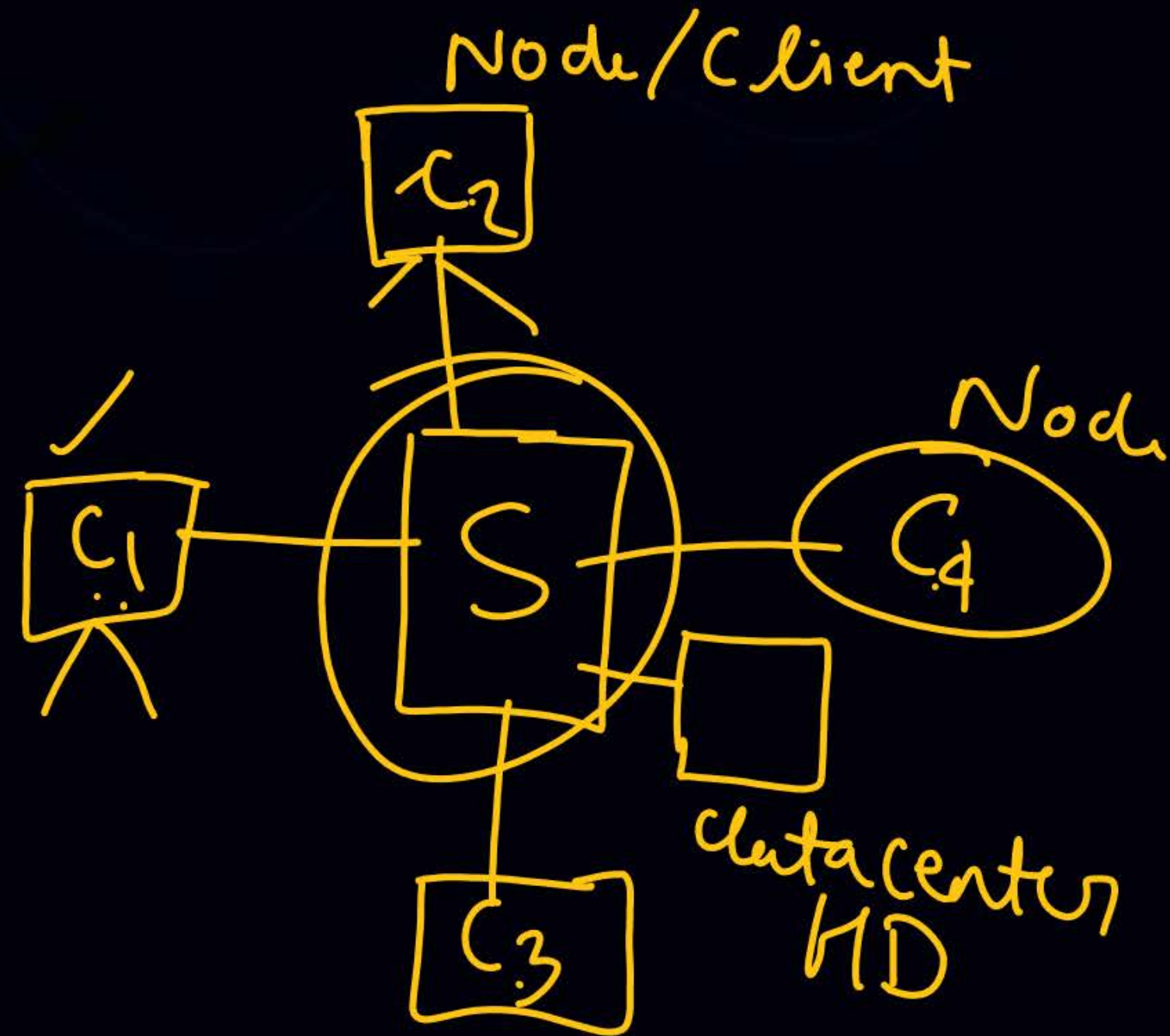
Topic : Database vs File System

Note:- Database is based on Client-Server Architecture.

1:- Fast Searching and Memory Utilization:- Searching is fast and Memory Utilization is less because of Client Server Architecture.

2:- Metadata:- No Metadata (Data about Data) is Required on where our data is stored.

3:- Concurrency :- Multiple Searching or Transaction done in same time, In DBMS there is PROTOCOL (RR, RW, WR, WW) to smooth work but In file System there is no any Protocol.

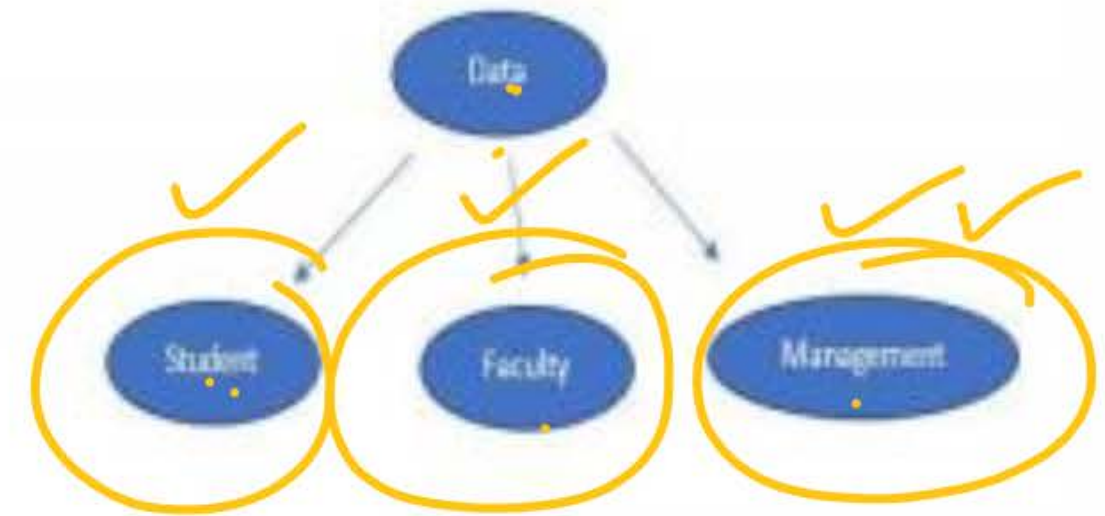


Topic : Database vs File System

Note:- Database is based on Client-Server Architecture.

1:- Security:- Roll based Access Control

Every Group get only Restricted data based on Roll Base.



2:- Redundancy:- Though the key Concept data Redundancy are solved via DBMS.

Patient Table

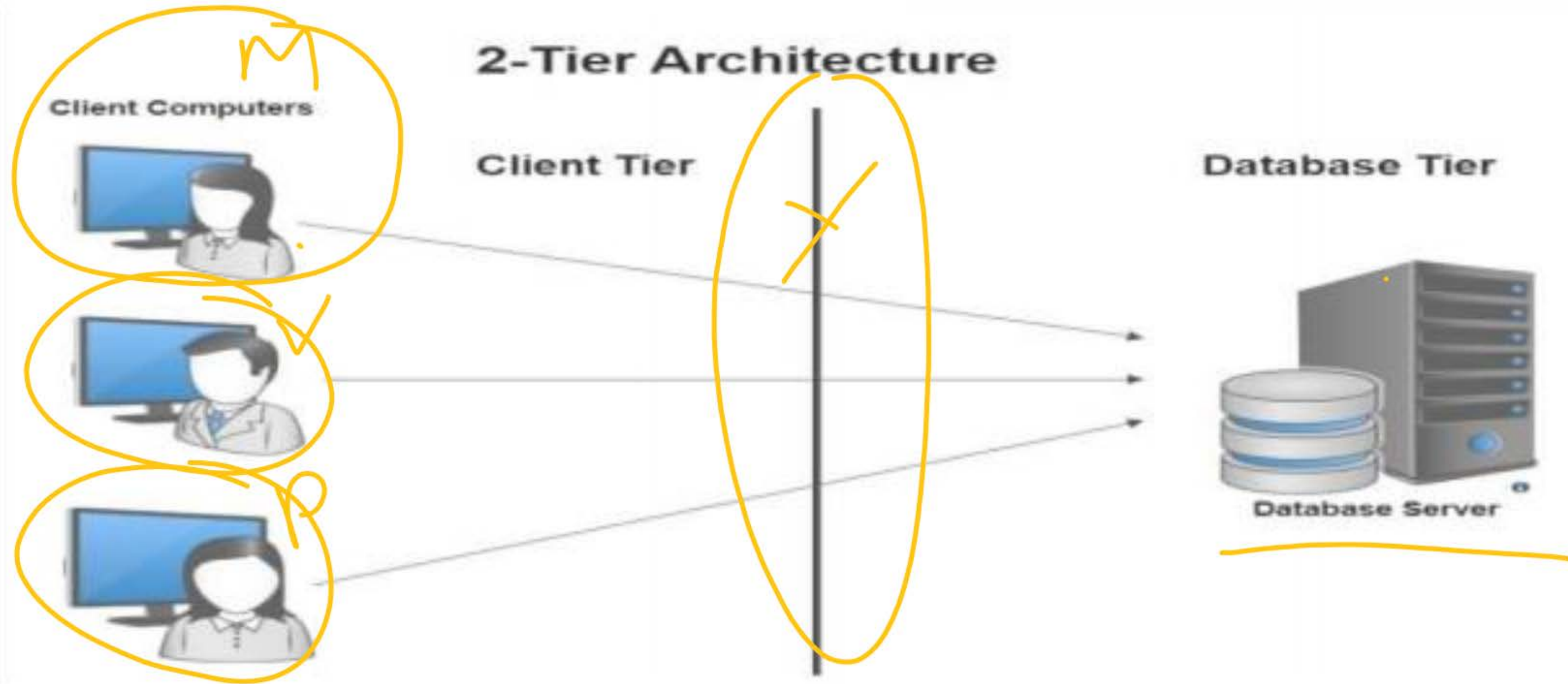
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258	Anu	8-Feb-1997	Female	8367242	02

Doctor Table

Doctor Id	Doctor	Visits	Room
01	Dr. Harpreet	7	03
02	Dr. Nishaat	9	06

Using the common field, one can read data from the other table.

Topic : 2 –Tier Architecture



Topic : 2 –Tier Architecture Advantage

Note:- Through JDBC via API or other API Client is directly Access to database Server.

EXAMPLE-1:- In Indian Railway you book ticket from Railway Counter Physically.

EXAMPLE-2:- In banks you Deposit or Withdraw money by Counter Physically.

Advantage:- Maintances Cost is less.

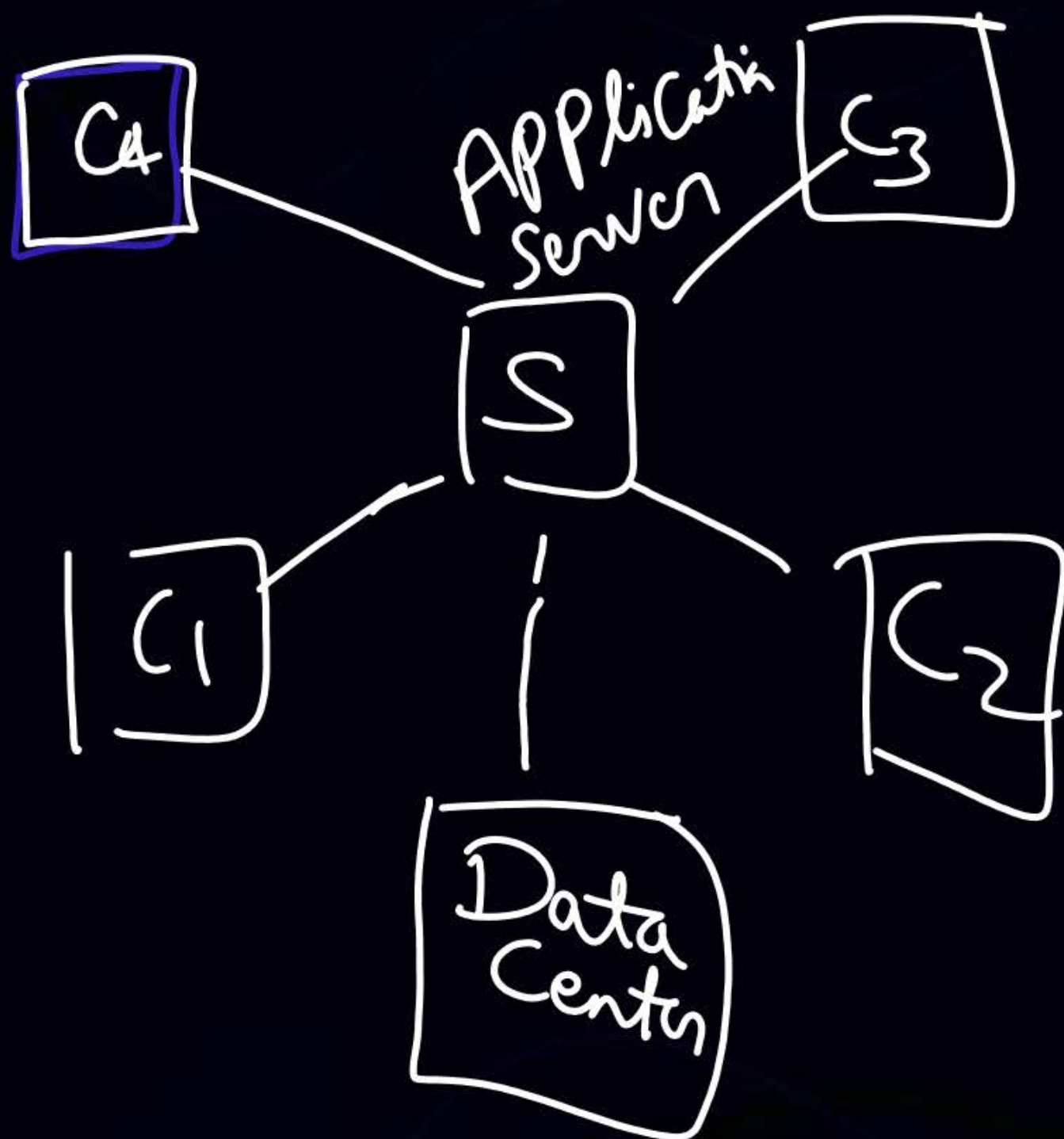
Topic : 2 –Tier Architecture Disadvantage

Disadvantage

Data Process (HLL $\rightarrow$ ML), Send and Receive the Data Packet all are doing by Database Server. On Database Server there is so many Loads.

1:- Scalability:- Due to Loads on Database Server we can not increase so much Client (Scaling). If Client will Increase so there is chance Database Server will Down.

2:- Security:- In 2-Layer Architecture, Client is directly access to Database Server which is not Secured.



Topic : 3-Tier Architecture Advantage

1:- **Scalability**:- we Solve the Scalability issue because of Application Server(Business Layer), Many Client Request process and manage by Application layer but data access by Database layer only.

2:- **Security**:- In 3-layer Architecture, Client is not directly Access to Database. In Business layer(Application Layer) there is so many Security patches are present.

Topic : RECAP

1:- DATABASE ✓

2:- DBMS and its types of DBMS- FLAT FILE, RDBMS

3:- Disadvantage of Flat File- Data Redundancy, Inconsistency, Security.

4:- Advantage of RDBMS OR DBMS- Searching fast, Concurrency solve, Security and Redundancy minimize.

5:- 2Tier Architecture

6:- Disadvantage and advantage of 2 Tier Architecture.

7:- 3Tier Architecture

8:- Advantage of 3 Tier Architecture.

THANK YOU